Raj Tandel

South San Francisco, CA | raj2511tandel@gmail.com | LinkedIn | GitHub

SUMMARY

Operations Research and Analytics professional with expertise in quantitative modeling, predictive analytics, and process optimization. Experienced in developing forecasting models, optimization solutions, decision-support systems, and data-driven operational workflows. Strong foundation in statistical analysis, simulation, and analytical techniques applied to operational planning, resource allocation, and performance evaluation. Proficient in Python, SQL, and analytical tools for building scalable quantitative solutions. Skilled in translating analytical results into actionable business insights, cross-functional collaboration, and project delivery in agile environments.

SKILLS

Productivity Tools:	MS Office 365, MS Excel, MS Project, JIRA, Power BI, Tableau, Trello, Asana, GitHub
Functional Skills:	Operations Research, Predictive Analytics, Statistical Modeling, Financial Forecasting, Optimization Techniques, Risk Modeling, Simulation, Decision Analysis, Scenario Analysis, Data Mining, Quantitative Problem-Solving, Process Improvement, Data Quality Monitoring
Software Skills:	Python (Pandas, NumPy, Scikit-Learn, SciPy, TensorFlow, PyTorch), SQL, R, MATLAB, FastAPI, Streamlit, Docker, API Integration
Soft Skills:	Agile (Scrum, Kanban), Risk Management, Resource Planning, Stakeholder Communication, Team Leadership, Cross-functional Collaboration, Strategic Problem Solving
Certifications:	Machine Learning by Stanford University (Coursera), Deep Learning Specialization by Andrew Ng (Coursera)

EXPERIENCE

All Air Mechanical Contractors Inc., South San Francisco, CA | Operations Research Analyst **May 2025 – Present**

- Develop and maintain quantitative models and decision-support systems spanning 10,000+ operational records, vendor datasets, and project documentation integrated with Trimble Viewpoint Spectrum ERP to support operational planning, scheduling, and resource allocation across field and office functions.
- Apply statistical analysis, predictive modeling, and analytical techniques to evaluate operational performance, cost drivers, and efficiency metrics across the company's business processes, reducing manual processing time from hours to seconds.
- Design and maintain decision-support tools and analytical systems to assist management with forecasting, workload balancing, and operational decision-making, converting unstructured business records into structured, analytics-ready datasets for reporting and performance evaluation.
- Analyze operational and financial data to identify process inefficiencies, optimization opportunities, and data-driven improvements supporting cost control and operational effectiveness across field operations, shop activities, and administrative functions.
- Translate analytical results into clear, actionable insights and recommendations for company leadership and cross-functional teams, supporting estimating, budgeting, pricing, and continuous improvement initiatives across the organization.

Inovatech Solutions, Gujarat, India | Data Scientist **March 2023 – August 2023**

- Developed predictive analytics and forecasting models for major manufacturing clients (L&T, Tata Motors), improving demand estimation accuracy and production planning efficiency through scalable statistical and analytical methods.
- Implemented automated quality inspection analytics for MG Motor India using pattern recognition and image analysis techniques, improving defect detection accuracy by 35% and reducing production rework rates across assembly and inspection stages.
- Designed predictive maintenance models for ABB India Ltd. using anomaly detection and time-series analysis, enabling early fault identification 48 hours in advance and reducing equipment downtime by 25% while optimizing asset reliability.
- Led a 6-member analytics team building data processing pipelines with Apache Spark and SQL, improving sensor data throughput and processing efficiency by 50% across manufacturing monitoring systems.
- Integrated real-time analytical workflows into industrial monitoring systems for adaptive operational decision-making across factory operations and production processes.

TechNova Analytics Solutions, Gujarat, India | Data Science Associate **August 2022 – March 2023**

- Designed and deployed end-to-end analytical models for finance and e-commerce applications, improving forecasting accuracy by 20% through advanced time-series modeling, regression analysis, and integration of macroeconomic trend data into predictive workflows.
- Built automated data processing pipelines using Python, Apache Spark, and SQL with version control, reducing preprocessing time by 30% and enabling real-time access to actionable business intelligence across stakeholder teams.
- Implemented customer segmentation and text-based analytics using clustering and classification techniques, boosting targeted marketing effectiveness and increasing customer satisfaction scores by 25%.
- Collaborated with cross-functional stakeholders to translate data insights into business strategies, while implementing data validation, quality monitoring, and performance checks for reliable analytical outputs.

Skyline Projects, Gujarat, India | Software Engineer Intern **January 2022 – August 2022**

- Developed web applications with integrated analytics dashboards and recommendation systems, increasing user engagement, retention, and personalized content delivery across customer-facing digital platforms.
- Integrated analytical models into responsive web platforms for real-time classification and anomaly detection, improving overall system efficiency by 15% and ensuring scalability under high traffic loads.
- Collaborated with cross-functional teams including UI/UX designers, backend developers, and data scientists, ensuring 100% on-time project delivery while maintaining clean, modular code and adhering to agile development best practices and CI/CD standards.

PROJECTS

Multimodal Data Curation for Conversational Systems | Vantech Solution | Canada **Spring 2025**

- Led curation of a multimodal labeled dataset integrating text, speech, and visual data to support advanced conversational system development across diverse use cases and demographics.
- Applied signal processing, contextual validation, and data augmentation techniques to ensure dataset robustness, reduce annotation bias, and enable reliable generalization across deployment scenarios.

Financial Risk Assessment and Portfolio Optimization Platform | Vester AI | USA **Spring 2025**

- Developed a financial risk assessment platform incorporating quantitative models for real-time market volatility analysis, economic forecasting, and investment risk evaluation.
- Implemented time-series forecasting and risk quantification techniques including Bayesian inference and macroeconomic analysis for proactive portfolio risk management and financial strategy optimization.

Investment Portfolio Optimization and Financial Analysis System | Northeastern University | USA **Spring 2024**

- Built an interactive financial analysis system integrating economic indicators, stock market data, and portfolio analytics to provide personalized investment recommendations based on user risk profiles.
- Developed an end-to-end portfolio optimization system using Markowitz Modern Portfolio Theory, Sharpe ratio analysis, and Monte Carlo simulations for data-driven asset allocation and risk assessment.

Stroke Prediction Model for Preventative Healthcare | Northeastern University | USA **Winter 2024**

- Built a predictive risk model using classification algorithms (Logistic Regression, Random Forest, Gradient Boosting), achieving an AUC of 0.86 for identifying high-risk patients for early intervention.
- Engineered features from 10,000+ patient records, refining critical predictors (hypertension, glucose levels, heart disease, and BMI) to enhance model interpretability and assist medical practitioners in designing early intervention and preventive care strategies.

EDUCATION

Northeastern University, Massachusetts, USA (GPA 3.9) **March 2025**
Master of Professional Studies in Applied Machine Intelligence, Concentration in AI with Finance

Gujarat Technological University, Gujarat, India (GPA 3.5) **June 2022**
Bachelor of Computer Engineering